Problema teoretica – set 3

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Grupa 231

3.17. Determinati aproximanta in sensul celor mai mici patrate

$$\varphi(t) = \frac{c_1}{1+t} + \frac{c_2}{(1+t)^2}, t \in [0,1]$$

a functiei $f(t) = e^{-t}$, luand $d\lambda(t) = dt$ pe $[0,1]$. Determinati numarul de conditionare $cond_\infty A = \|A\|_\infty \|A^{-1}\|_\infty$ al matricei A a coeficientilor ecuatiilor normale. Calculati eroarea $f(t) - \varphi(t)$ in $t = 0, t = \frac{1}{2}$ si $t = 1$.

(Indicatie: Integrala $\int_1^\infty t^{-m} e^{-xt} dt = E_m(x) = E_i(m,x)$ se numeste a "m-a integral exponentiala". Exprimati rezultatul cu ajutorul acestei functii.)

Rezolvare:

1) Se doreste ca $S = \|\varphi(t) - f(t)\|$ sa fie minim. Obtinerea valorilor $c_1 si\ c_2$ se face din sistemul

$$\begin{cases} \dfrac{\partial S}{\partial c_1} = 0 \\ \dfrac{\partial S}{\partial c_2} = 0 \end{cases}$$

$$S = \int_0^1 \left(\frac{c_1}{1+t} + \frac{c_2}{(1+t)^2} - e^{-t}\right)^2 dt$$

$$\frac{\partial S}{\partial c_1} = 0 \Rightarrow \int_0^1 2\left(\frac{c1}{1+t} + \frac{c_2}{(1+t)^2} - e^{-t}\right)\frac{1}{1+t}\ dt = 0$$

$$\frac{\partial S}{\partial c_2} = 0 \Rightarrow \int_0^1 2\left(\frac{c_1}{1+t} + \frac{c_2}{(1+t)^2} - e^{-t}\right)\frac{1}{(1+t)^2}\ dt = 0$$

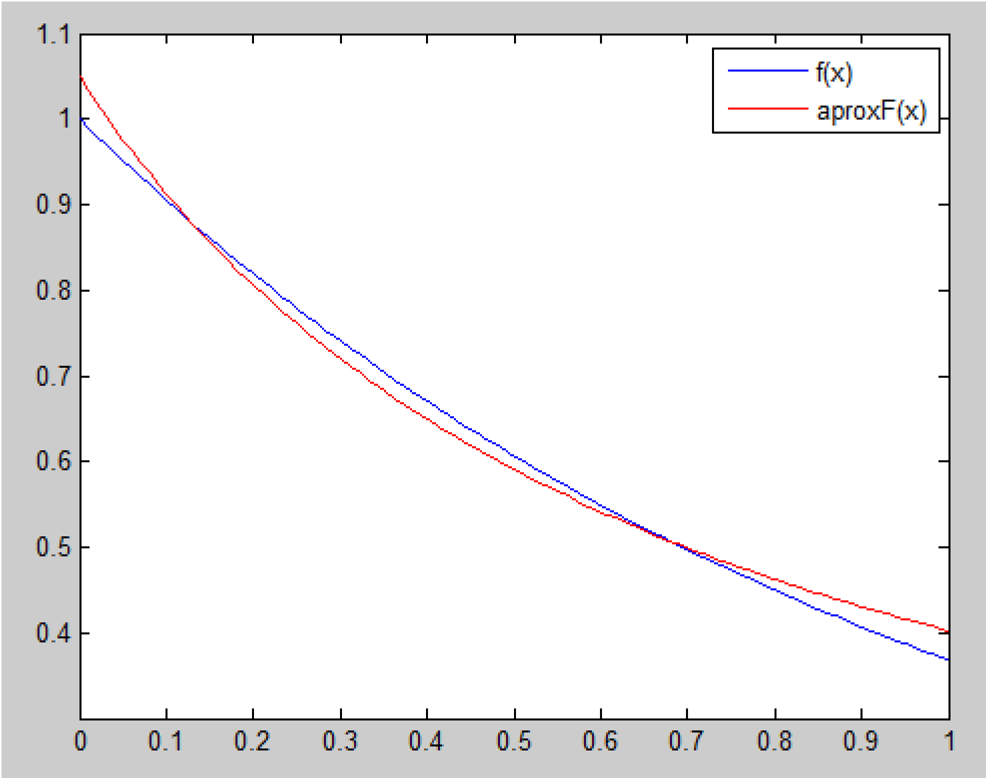
$$c_1 \int_0^1 \frac{dt}{(1+t)^2} + c_2 \int_0^1 \frac{dt}{(1+t)^3} = \int_0^1 \frac{e^{-t}}{1+t} dt$$

$$c_1 \int_0^1 \frac{dt}{(1+t)^3} + c_2 \int_0^1 \frac{dt}{(1+t)^4} = \int_0^1 \frac{e^{-t}}{(1+t)^2} dt$$

Introducand in matlab ecuatiile de mai sus obtinem c1 = 0.5616959999999995 si c2 = 0.4868640000000006. Functia aproximanta este

$$\varphi(t) = \frac{0.561695999999999}{1+t} + \frac{0.4868640000000006}{(1+t)^2}, t \in [0,1]$$

O reprezentare vizuala a aproximantei este urmatoarea:



2) Matricea A a coeficientilor ecuatiilor normale:

$$A = \begin{pmatrix} \int_0^1 \frac{dt}{(1+t)^2} & \int_0^1 \frac{dt}{(1+t)^3} \\ \int_0^1 \frac{dt}{(1+t)^3} & \int_0^1 \frac{dt}{(1+t)^4} \end{pmatrix}$$

3) Calculati eroarea $f(t) - \varphi(t)$ in $t = 0, t = \frac{1}{2}$ si $t = 1$.

$$\varphi(t) = \frac{0.561695999999999}{1+t} + \frac{0.4868640000000006}{(1+t)^2}, t \in [0,1]$$

$$f(0) - \varphi(0) = 1 - 1.04856 = -0.0485600000000002$$

$$f\left(\frac{1}{2}\right) - \varphi\left(\frac{1}{2}\right) = 0.6065 - 0.590848 = 0.015652$$

$$f(1) - \varphi(1) = 0.3679 - 0.402563999999999 = -0.034663999999999$$